📄 AI Tools Project – Knowledge Transfer (KT) Document

Author: Anna Swetha & Indhuja
Role: Associate Consultant – Technology
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1. Introduction & Purpose

The **AI Tools Explorer** is a web-based application designed to help users **discover, explore, and learn about AI tools**. It uses an **interactive D3.js Sunburst Chart**, a **search bar**, and **video demos** to make exploration easy.

This document serves as a **Knowledge Transfer (KT) guide** so that any new developer or team member can understand:

- The project's purpose and architecture.
- How the system works end-to-end.
- How to deploy, run, and maintain the project.

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2. Project Overview

The **AI Tools Explorer** allows users to:

- **Visually explore** AI tool categories using a **Sunburst Chart**.
- **Search** for a specific tool or category.
- **View tool details** in a modal popup.
- **Watch demo videos** related to tools.

Example:

- Live Demo: AI Tools Explorer
- Repository: GitHub Link

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3. Tech Stack

Layer	Technology
Frontend	Angular 16+, HTML, CSS, TypeScript, D3.js
Backend	Node.js, Express.js
Database	JSON-based data (in backend)
Deployment	Vercel (frontend), Docker + Google Cloud Run (backend)

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4. Folder Structure

```
project-root/
│
├── server.js       # Main backend entry file
├── package.json     # Backend dependencies
├── Dockerfile      # Backend Docker configuration
│
├── frontend/
│   ├── src/
│   │   ├── app/
│   │   │   ├── components/
│   │   │   │   ├── search-bar/      # Search functionality
│   │   │   │   ├── sunburst-chart/    # D3.js chart logic
│   │   │   │   ├── tool-info-display/  # Tool details & modal
│   │   │   ├── services/          # API calls
│   │   │   └── app.module.ts
│   │   └── index.html
│   ├── angular.json
```

```
│   ├── package.json
│
└── README.md
```

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5. Detailed Features & Workflows

5.1 Search Function

Purpose: Quickly find categories or tools.

Workflow:

1. The user types a term in the **Search Bar**.
2. Frontend sends a request: `GET /api/search?q=<term>`
3. Backend returns matching categories or tools.
4. If category found → chart **zooms** to category.
5. If tool found → chart **highlights** tool + opens detail modal.

Code Reference:

- Frontend: `search-bar.component.ts`
- Backend: `/api/search` route in `server.js`

5.2 Tools Display

Purpose: Show all tools in a selected category.

Workflow:

1. User clicks on category in **Sunburst Chart**.

2. Frontend sends request:

   GET /api/categories/:id/tools

3. Backend returns tools for that category.
4. Chart zooms into that category.
5. Tools are displayed → clicking a tool opens detail modal.

5.3 Video Demo

Purpose: Show video demo for a tool.

Workflow:

1. The user clicks "Play Video" in tool detail.
2. Frontend sends request:GET /api/videos/:toolId
3. Backend returns video URL and metadata.
4. Video plays in embedded player.

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6. Backend API Details

Endpoint	Method	Description
/api/categories	GET	Get all categories
/api/categories/:id	GET	Get category details
/api/categories/:id/tools	GET	Get tools in category
/api/tool/:id	GET	Get tool details

/api/search?q=term	GET	Search categories or tools
/api/videos/:toolId	GET	Get video details
/api/videos/search?q=term	GET	Search videos

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7. Deployment Procedure

7.1 Frontend – Vercel

1. Push Angular code to GitHub.
2. Open Vercel Dashboard → **New Project**.
3. Link GitHub repo.
4. Configure:
    - Framework: **Angular**
    - Build Command: ng build --prod
    - Output Directory: dist/<project-name>
5. Deploy → get live URL.

7.2 Backend – Docker + Google Cloud Run

Step 1: Build Docker Image

```
docker build -t ai-tools-backend .
```

Step 2: Test Locally

```
docker run -p 8080:8080 ai-tools-backend
```

Step 3: Authenticate with Google Cloud

```
gcloud auth login
gcloud config set project YOUR_PROJECT_ID
gcloud auth configure-docker
```

Step 4: Push Image to GCR

```
docker build -t gcr.io/YOUR_PROJECT_ID/ai-tools-backend:latest .
docker push gcr.io/YOUR_PROJECT_ID/ai-tools-backend:latest
```

Step 5: Deploy to Cloud Run

```
gcloud run deploy ai-tools-backend --image
gcr.io/ai-tools-project-468011/ai-tools-backend:latest --platform managed --region
us-central1 --allow-unauthenticated
```

Step 6: Update Angular environment file with backend URL.

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8. Local Development Setup

Frontend:

```
cd frontend
npm install
ng serve
```

```
# http://localhost:4200
```

Backend:

```
npm install
node server.js
# http://localhost:8080
```

---

9. Workflow Diagram

```
[User]
 ↓
[Search in Search Bar]
 ↓ calls
/api/search → Returns matching categories/tools
 ↓
If Category → Zoom Sunburst → Show tools
If Tool → Highlight → Show Tool Details
 ↓
User clicks "Play Video"
 ↓ calls
/api/videos/:toolId → Returns video URL
 ↓
Video plays in embedded player
```

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10. Common Issues & Fixes

- **Chart not zooming:** Ensure D3.js event binding is correct.
- **Video not loading:** Check `/api/videos/:toolId` response.

- **CORS error:** Enable CORS in backend server.js.

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11. Future Enhancements

- Add real database (MongoDB or PostgreSQL).
- User login & favorites.
- Tool ratings and reviews.
- Category-level analytics.

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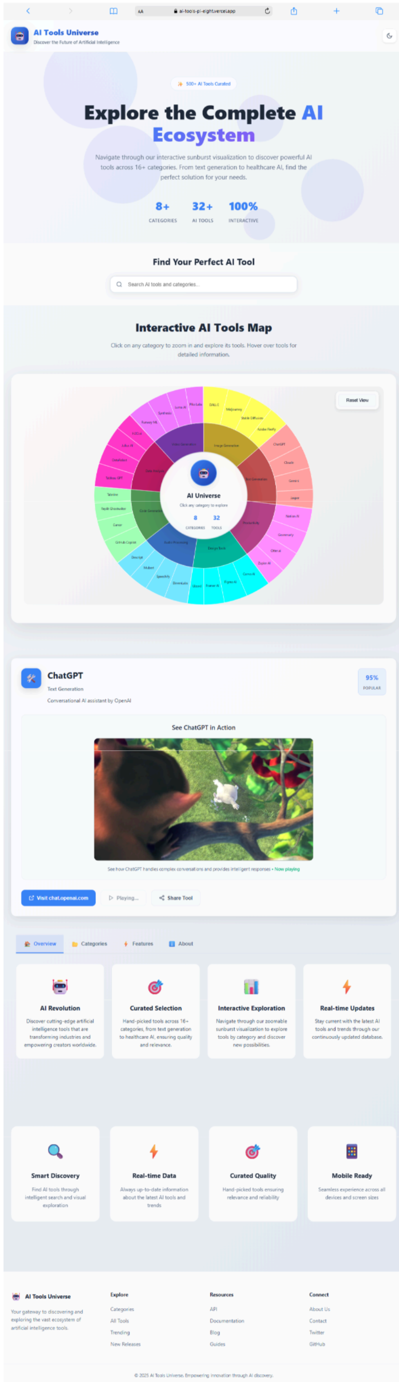
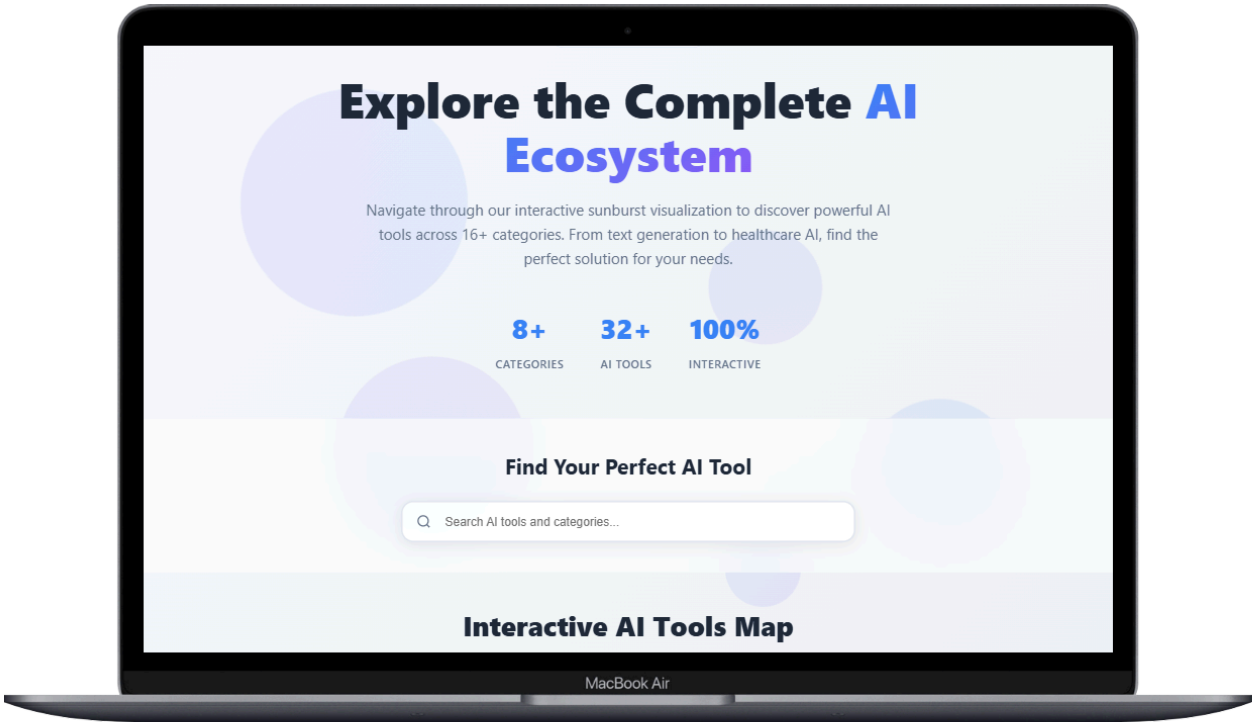
12. Screenshot Placeholders

- *[Homepage Screenshot Here]*



- *[Search Bar Screenshot Here]*



- *[Sunburst Chart Zoom Screenshot Here]*

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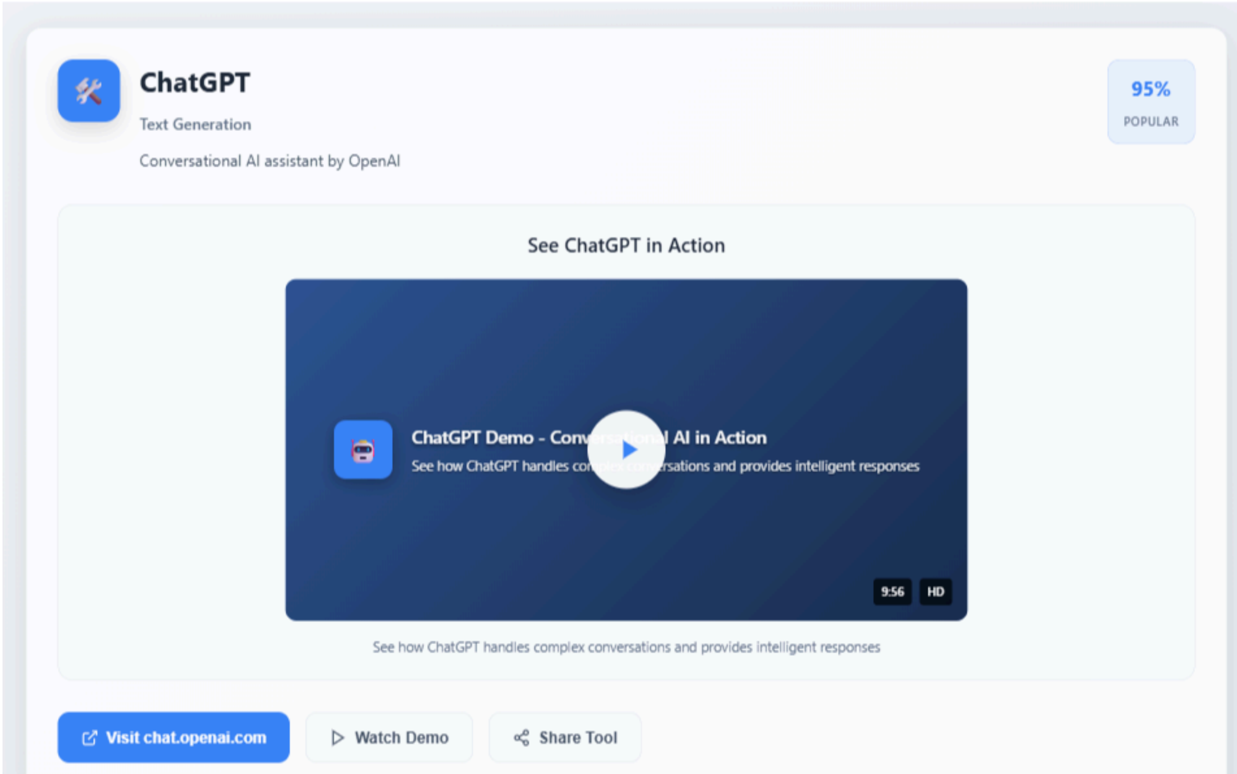
- *[Tool Detail Modal Screenshot Here]*



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- *[Video Playback Screenshot Here]*